Ringmo-SenseV2: Remote Sensing Foundation Model for Spatiotemporal Prediction Based on Multisource Heterogeneous Time-Series Data

Liangyu Xu, Wanxuan Lu, Leiyi Hu, Heming Yang, Yi Jiang, Chenglong Liu, Hongfeng Yu, Chubo Deng, Xian Sun, *Senior Member, IEEE*, and Kun Fu, *Senior Member, IEEE*

***Abstract*— The rapid development of remote sensing (RS) technology has generated a vast amount of heterogeneous time-series data from various sources, including drone videos, satellite time-series images, and multiobject trajectories. Effectively processing and analyzing this multisource heterogeneous data for accurate spatiotemporal prediction is crucial in fields such as environmental protection and disaster response. In this article, we propose a universal predictive foundation model named Ringmo-SenseV2 to learn the general evolutionary patterns of RS elements from massive heterogeneous data. Ringmo-SenseV2 features a mixture-of-heterogeneous-experts (MoHE) Transformer, which unifies the modeling of multisource heterogeneous time-series data. Additionally, to better capture the complex dependencies across different spatiotemporal locations, we introduce a hypergraph translator (HT), treating embeddings of different spatiotemporal locations as nodes and employing hypergraph convolution for information propagation. Furthermore, to enhance the model's adaptability to different evolution speeds during pretraining, we implement the adaptive tube masking (AM) strategy, which controls prediction difficulty by adaptively setting mask proportions for sequences with varying evolution speeds. Extensive experiments demonstrate that Ringmo-SenseV2 exhibits outstanding performance across various RS prediction tasks. Further tests on scene graph generation for RS images showcase the model's ability to extract image features, thereby enhancing image perception tasks.**

***Index Terms*— Foundation model, remote sensing (RS), spatiotemporal prediction.**

Received 21 August 2024; revised 13 January 2025; accepted 27 March 2025. Date of publication 28 April 2025; date of current version 13 May 2025. This work was supported by the National Key Research and Development Program of China under Grant 2021YFB3900504. *(Liangyu Xu and Wanxuan Lu contributed equally to this work.) (Corresponding author: Wanxuan Lu.)*

Liangyu Xu, Leiyi Hu, Heming Yang, Yi Jiang, Chenglong Liu, Xian Sun, and Kun Fu are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, also with the Key Laboratory of Target Cognition and Application Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: xuliangyu21@mails.ucas.ac.cn; huleiyi21@mails.ucas.ac.cn; yangheming21@mails.ucas.ac.cn; jiangyi22@mails.ucas.ac.cn; liuchenglong20@mails.ucas.ac.cn; sunxian@aircas.ac.cn; kunfuiecas@gmail.com).

Wanxuan Lu, Hongfeng Yu, and Chubo Deng are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the Key Laboratory of Target Cognition and Application Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China (e-mail: luwx@aircas.ac.cn; yuhf@aircas.ac.cn; dengcb@aircas.ac.cn).

Digital Object Identifier 10.1109/TGRS.2025.3565203

I. Introduction

As TIME progresses, objects on Earth are constantly changing. Understanding their spatiotemporal evolution patterns and accurately predicting future trends hold great potential for various industries, including traffic management, environmental conservation, and disaster response. With the rapid development of remote sensing (RS) technology, Earth observation can achieve global coverage with all-weather and full-spectrum detection capabilities. Daily data collection spans hundreds of millions of square kilometers worldwide, characterized by vast data volumes, complex scenes, rich content, and diverse temporal phases. The extensive and diverse data provide strong support for predicting the evolution of RS elements.

Nevertheless, existing prediction methods are largely designed for specific type of data, such as trajectory prediction [1], [2], [3], cloud forecast [4], and Earth surface prediction [5]. While these methods perform well in their respective scenarios, they have limited generalization capabilities and cannot integrate diverse heterogeneous data to make more accurate predictions. Therefore, there is an urgent need for a more general and broadly applicable solution.

The "Pretraining + Finetuning" paradigm has proven to be an effective approach. Several RS foundation models have gradually emerged, demonstrating satisfying performance in various tasks [6], [7]. These models have primarily focused on perception tasks such as recognition, detection, segmentation, and change detection. Recent research has also applied the foundation model to tasks with RS time-series data, including RS video prediction, precipitation forecasting, object tracking, and RS segmentation, and has achieved remarkable results [8]. However, when faced with heterogeneous data, existing models suffer from a lack of generalization and versatility, as a single model cannot simultaneously process point data (trajectories) and spatial data (time-series images and video frames).

Therefore, we attempt to construct a universal predictive foundation model for heterogeneous time-series RS data, to uncover the temporal evolution patterns and support multiscenario prediction tasks. To solve this issue, we primarily face three challenges, as shown in Fig. 1.

1) *Data Heterogeneity:* Earth observation data come from a wide range of sources and exhibit significant variability

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

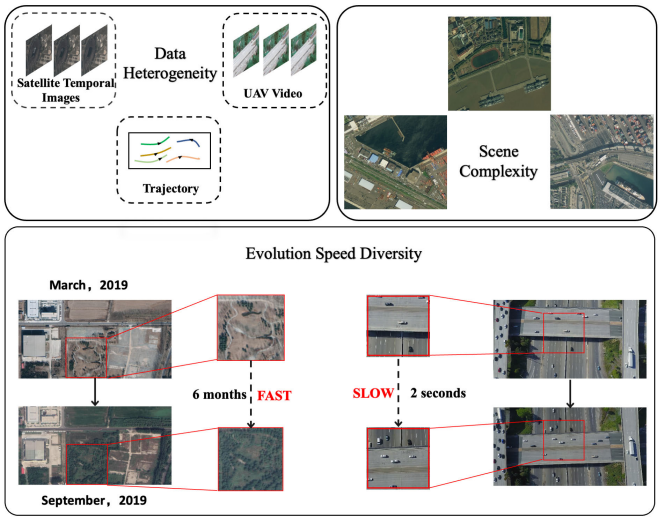


Fig. 1. Characteristics of time-series RS data: data heterogeneity, scene complexity, and evolution speed diversity.

in form and dimension. For instance, trajectories are point sequences, while time-series images and videos are spatial sequences. Unified modeling of these heterogeneous RS temporal data types presents considerable challenges.

2) *Scene Complexity:* RS scenes encompass vast spatial scales, with a multitude of diverse objects scattered extensively. They do not exist in isolation but have various connections with each other. It is difficult to explicitly model complex dependencies among various locations to enhance the prediction of object movements and overall scene changes.
3) *Evolution Speed Diversity:* The evolution speeds of spatiotemporal data vary across geographical regions and environmental conditions. Changes between adjacent satellite images can be significant, while differences between adjacent unmanned aerial vehicle (UAV) video frames are relatively small. This leads to varying predictive difficulties. During pretraining, the model converges more quickly on simpler samples, hindering its ability to learn features of complex samples and reducing generalization performance in downstream tasks.

To address these issues, we propose a universal predictive foundation model framework for heterogeneous RS time-series data, named Ringmo-SenseV2. It consists of two components: a large-scale RS time-series dataset with heterogeneous data from various sources and a foundational model for RS prediction tasks. In order to learn the general evolution patterns of different RS elements, we have gathered a dataset that includes UAV videos, satellite temporal images, paired RS video–trajectory sequences, and trajectories for ships. We further perform distinct preprocessing procedures for each data type. Additionally, to unify the modeling of heterogeneous time-series RS data, we construct a general predictive foundational model for RS time-series data. The model facilitates the shared learning of multidomain heterogeneous multitemporal data, extracting stable and generalized temporal evolution features, and supports prediction tasks across multiple spatial and temporal scales. Our model comprises a mixture-of-heterogeneous-experts (MoHE) Transformer encoder, a hypergraph translator (HT), and two decoders for videos and trajectories. Video/temporal images, trajectories, and video–trajectory pairs serve as the three types of inputs during the pretraining phase. Different types of data share the attention mechanism while having distinct feedforward layers, designed as modal experts within MoHE. The HT, situated between the encoder and decoders, models intricate relationships among different spatiotemporal locations. Moreover, in order to address the diversity in evolution speed, we design an evolution speed factor to precisely measure the evolution speed of different sequences. Based on the distribution of data evolution speeds, we set different masking rates for time-series sequences, balancing the prediction difficulty of different samples to improve the efficiency of pretraining.

Based on the collected dataset, we apply masked data modeling (MDM) on the foundation model and the pretrained backbone can be applied to a wide range of RS downstream prediction tasks. We validate the proposed framework's effectiveness on six downstream tasks: Earth surface forecasting, RS video prediction, RS segmentation forecasting, ship trajectory prediction, RS video–trajectory joint prediction, and RS scene graph generation. The experimental results demonstrate that our model has achieved competitive results across multiple tasks. In summary, the contributions of this article are as follows.

1) We introduce Ringmo-SenseV2, a universal RS spatiotemporal predictive foundation model framework, which consists of a large-scale time-series RS dataset with heterogeneous data from various sources, and a foundational model for RS prediction tasks.
2) To extract stable and generalized spatiotemporal features, we propose a general foundation model that uses an MoHE encoder to unify the modeling of different modalities and incorporates HT for complex dependency modeling of spatiotemporal location information, establishing an information pathway for multisource heterogeneous time-series data input and diversified task output.
3) To handle diverse evolution speeds, we introduce an evolution speed factor for precise measurement across sequences and propose the adaptive tube masking (AM) strategy, balancing the prediction difficulty of different samples and thereby improving the efficiency of pretraining.
4) We pretrain the foundation model on the collected dataset and fine-tune it on various downstream tasks. Highly competitive results demonstrate Ringmo-SenseV2's adeptness in understanding the spatiotemporal evolution patterns of RS elements and its robust generalization capability.

II. Related Work

This section primarily introduces research related to RS time-series prediction and foundational models. Section II-A

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

reviews existing RS spatiotemporal prediction tasks and classical methods. Section II-B introduces foundation models in the RS field, outlining their basic methods and the problems they have addressed.

A. Predictive Learning in RS

Predictive learning tasks in RS can be categorized based on the input data type, primarily encompassing spatiotemporal prediction (for raster data) and trajectory prediction (for point data). Spatiotemporal prediction encompasses tasks such as precipitation nowcasting, cloud forecast, Earth surface prediction, and video prediction. In recent years, with the flourishing development of deep learning, a plethora of works have emerged to address these problems, achieving outstanding results [9], [10], [11], [12], [13]. These methods are based on recurrent neural networks (RNNs) and modify long short-term memory (LSTM) to adapt to time-series sequence data, addressing issues such as gradient vanishing, historical information forgetting, and long-term decay. They have achieved remarkable results in spatiotemporal prediction tasks.

Trajectory prediction encompasses ship, typhoon, and vehicle trajectories in RS scenes. Methods can be broadly categorized into traditional statistical probability approaches and neural network models. Statistical probability methods include Markov chains and Bayesian models, which generally consider the motion trajectories of different entities collectively. In recent years, the development of deep learning has provided new perspectives for trajectory prediction. RNNs and their variants, such as LSTM [14], have been widely applied in trajectory prediction [15]. The emergence of the Transformer architecture [16] has sparked further research in trajectory prediction [17], [18], [19], [20], [21], [22]. Leveraging the Transformer's ability to model long-range dependencies, these methods have pushed trajectory prediction accuracy to a higher level.

B. Foundation Models in RS

The strong performance of foundation models in tasks such as image classification, object detection, and semantic segmentation for natural scenes [23], [24], [25] has sparked researchers' interest in RS foundation models [6], [7], [26]. An ordinary Vision Transformer (ViT) model with around one million parameters is employed, marking the first effort to introduce a foundation model designed specifically for RS tasks [7]. They design a rotation-variable-size window attention mechanism to address the characteristics of large-scale and arbitrarily oriented objects in RS images. This significantly reduced computational costs and memory usage while achieving state-of-the-art (SOTA) results in detection tasks and competitive results in classification and segmentation tasks. Ringmo [6] is a generative pretrained model designed for RS data, deeply integrating the imaging mechanisms and object characteristics of RS data. It has achieved SOTA performance in typical downstream tasks such as scene classification, object detection and localization, fine-grained recognition, feature extraction, and change detection. SatMAE [26] is a pretraining framework designed for time-series or multispectral satellite images. It incorporates a masked autoencoder approach, featuring time embeddings and independently masked image blocks across different time points. This framework has showcased significant enhancements in both supervised learning performance on benchmark datasets and transfer learning performance on various downstream RS tasks, such as land use classification and semantic segmentation. Recent research RingMo-Sense has also applied the foundation model to RS time-series tasks, including RS video prediction, precipitation forecasting, object tracking, and RS segmentation, and has achieved remarkable results [8].

While these efforts have significantly advanced perceptual tasks in the RS domain, there is currently no research attempting to establish a foundational prediction model to provide a universal solution for RS heterogeneous data.

III. RINGMO-SENSEV2 FRAMEWORK

In this article, we propose Ringmo-SenseV2, a universal predictive foundation model framework for RS predictive learning, aiming at training a foundation model on the large-scale collected dataset to learn the general evolution patterns of multisource heterogeneous RS time-series data, which can then be applied to numerous downstream tasks. Section III-A introduces the dataset collected for pretraining, and Section III-B elaborates on key designs of the foundation model.

A. Dataset

The evolution of RS elements is influenced by environmental conditions, geographical location, and climate change, presenting complex and varied patterns. To ensure that the foundation model fully explores the evolution patterns of Earth elements, the dataset for pretraining must possess a high degree of diversity.

1) Dataset Collection and Preprocess: To effectively pretrain the foundation model, we have gathered massive RS time-series data from multiple sources, as shown in Table I and Fig. 2. The dataset comprises four types of data: videos, temporal images, trajectories, and video–trajectory pairs. For each data type, we perform distinct preprocessing procedures.

a) Videos/satellite temporal images: Videos and satellite temporal images from various sources have different storage formats, sequence lengths, and resolutions. We create a standardized dataset by keeping .mp4 format data unchanged and converting image sequences into .mp4 format by dividing them into 300-frame segments. Additionally, to ensure that the final data input into the model has a consistent resolution, for sequences of varying resolutions, we randomly crop sequences of 512×512 sized images and scale them to 224×224 as the model input. Moreover, for videos and satellite temporal images, we incorporate data augmentation techniques such as random horizontal flipping (with a 50% probability) and brightness adjustment (randomly varying within the range [0.8, 1.2]). These augmentations enhance the model's robustness and increase the diversity of the training data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

TABLE I

STATISTICAL INFORMATION OF THE COLLECTED DATASETS FOR PRETRAINING THE FOUNDATION MODEL. VT-PAIR MEANS PAIRED DATA OF VIDEO AND TRAJECTORY. AIS STANDS FOR AUTOMATIC IDENTIFICATION SYSTEM

Dataset	Type	Platform	Frames/Trajectory Number	Resolution	Categories
AU-AIR[28]	Video	UAV	32823	1920×1080	8
DroneCrowd[29]	VT-Pair	UAV	33600	1920×1080	1
VisDrone[30]	Video/VT-Pair	UAV	261908	3840×2160	10
VISO[31]	VT-Pair	Satellite	15455	12000×5000	4
MDMT[32]	VT-Pair	UAV	39678	1920×1080	3
AnimalDrone[33]	Video	UAV	53644	3840×2080	1
Anti-UAV[34]	Video	UAV	723000	640×512	1
UAVDT[35]	Video	UAV	8000	1080×540	3
Stanford Drone Dataset[36]	VT-Pair	UAV	929499	1424×1088	6
UCLA Aerial Event Dataset[37]	Video	UAV	27 videos	2386×1723	12
WebUAV-3M[38]	VT-Pair	UAV	3.3 million	608×342	223
SatSOT[39]	VT-Pair	Satellite	27664	12000×5000	4
Sea Drones See[40]	VT-Pair	UAV	54000	3840×2160	1
Small Aircraft Target Detection[41]	VT-Pair	UAV	16177	1920×1080	1
ETH/UCY[42, 43]	Trajectory	-	2206	-	-
Argoverse Motion Forecasting Dataset[44]	Trajectory	-	5000	-	-
EarthNet2021[45]	Temporal images	Satelite	2269 sequences	128×128	-
Self-built data	VT-Pair	Satellite	397 videos	1920× 1080	2
Self-built data	Trajectory	AIS	5000	-	-

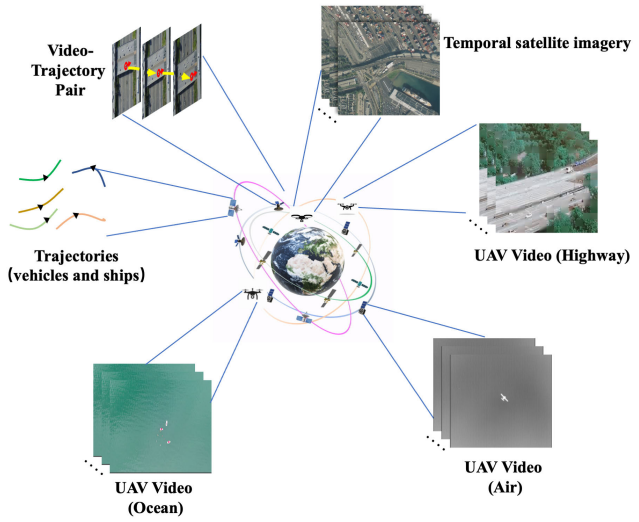


Fig. 2. Dataset we collect comprises heterogeneous time-series RS data from various sources, including drone videos and satellite time-series images representing different scenes, as well as video–trajectory pairs and trajectory data.

b) Trajectories: There are two different coordinate systems where the trajectories are represented in the collected dataset: the pixel coordinate system and the world coordinate system. To ensure that our model can smoothly handle trajectories coming from these two different coordinate systems, we perform a two-step normalization process. First, we individually normalize trajectories within each coordinate system, which ensures that trajectory data within their respective coordinate systems have the same scale and distribution centered around the mean. In the second step, we perform global standardization. We restandardize the trajectories obtained from the first step. This further ensures that the data are standardized, allowing trajectories from both the pixel coordinate system and the world coordinate system to share the same reference frame and scale. The final standardized trajectories are used as input for the model and are augmented with zero-mean Gaussian noise (a standard deviation of 0.01). This noise perturbation simulates potential measurement errors and improves the model's ability to handle uncertainty.

c) Video–trajectory Pairs: The preprocessing for videos in video–trajectory pairs is the same as described above for videos. Trajectories in the paired data are represented in pixel coordinates, so they require only a single-step standardization.

2) Statistical Information: Following the preprocessing steps described in Section III-A1, we obtain 2269 satellite temporal images, 20 466 UAV videos, 12 182 video–trajectory pairs, and 59 594 trajectories. The dataset encompasses diversity and richness. These videos capture a range of environments, including urban areas, grasslands, and mountainous regions, offering diverse aerial perspectives. As for trajectory data, it boasts a substantial volume, featuring 19 398 vehicle trajectories, 35 196 pedestrian trajectories, and 5000 ship trajectories from automatic identification system (AIS). This amalgamation of diversity and quantity provides robust data support for our foundational model, enabling it to gain a deeper understanding of the evolution patterns of different scenarios and object trajectories.

3) Temporal Redundancy Analysis: Time-series data typically exhibit temporal correlations, meaning that different frames in time-series images or videos often have similarities. This similarity leads to information redundancy, as much of the content in adjacent frames is repetitive or highly correlated. This redundancy is evident not only in image sequences but also in video data. Since each frame captures some content from the previous frame, the data contain a significant amount of duplicate information. Identifying and handling this redundant information is a crucial aspect of time-series data analysis, which helps improve the efficiency and accuracy of models.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

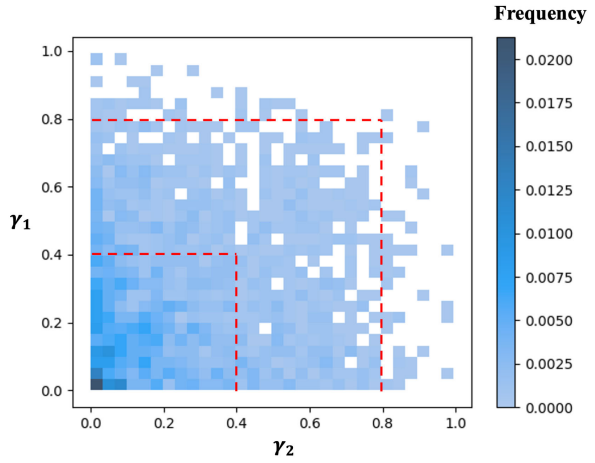


Fig. 3. Temporal redundancy distribution. For each clip, we obtain two sequences and calculate their temporal redundancy factors, denoted as γ_1 and γ_2.

We measure temporal redundancy by designing a factor γ, based on the structural similarity index (SSIM) between images [45]. For any clip $\boldsymbol{X}$, we start at any possible position and sample T frames at intervals of τ, denoted as $x_1, x_2, \ldots, x_T$. Then, γ is calculated as follows:

$$\gamma(\boldsymbol{X}) = \frac{S(x_1, x_T) + S(x_1, x_m) + S(x_m, x_T)}{3} \tag{1}$$

where $m = \lfloor (1+T)/2 \rfloor$, $S(x_i, x_j)$ represents the SSIM between the ith frame and the jth frame. We take $T = 16$ and $\tau = 4$ and calculate the temporal redundancy of all time-series data in this dataset. To avoid sampling randomness, we sample twice starting from random positions each time. In other words, for each clip, we obtain two sequences and calculate their temporal redundancy factors, denoted as γ_1 and γ_2. The 2-D distribution histogram is shown in Fig. 3. The temporal redundancy of time-series data is distributed across various intervals. Most sequences exhibit low temporal redundancy, while a substantial portion of sequences has a temporal redundancy factor γ ranging from 0.4 to 0.8. Only a few sequences have temporal redundancy exceeding 0.8.

Temporal redundancy has a significant impact on the difficulty of MDM in the foundation model. Specifically, temporal redundancy quantifies the degree of frame similarity, which is inversely correlated with evolution speed, a measure of the rate of temporal changes in the scene. If not handled appropriately, temporal redundancy can lead to a decrease in the model's generalization ability because the model might overly rely on repeatedly occurring information and ignore more valuable variations. On the other hand, temporal redundancy also presents an opportunity for the model. It allows the model to capture and understand the evolution patterns of objects at different time scales. By comparing and analyzing data at different time steps, the model can make better predictions about the future states of the objects. Therefore, effectively utilizing temporal redundancy can enhance the performance of the base model. However, this requires careful modeling and handling to strike a balance between redundancy and information richness.

B. Foundation Model Architecture

To address the issues described in Section I, we customize the design of the foundation model architecture, as illustrated in Fig. 4. The input to the foundation model includes one video clip (sampled every T frames with a stride of τ from the original video), one trajectory (consisting of T trajectory points), and one video–trajectory pair (containing video frames and trajectory points at T time steps). It is worth noting that in the data path, we treat satellite time-series images and drone videos equally based on their data formats. We conduct pretraining for Ringmo-SenseV2 by unified masked data modeling (MDM) [46] with an asymmetric encoder-decoder architecture on both homogeneous data (videos and trajectories) and heterogeneous data (video–trajectory pairs). For video input, since we have already unified the processing of data from different sources, we directly employ spatiotemporal cube embedding [25], [47], [48], where we treat each cube of size $2 \times 16 \times 16$ as a token embedding. Therefore, the cube embedding layer generates $(T/2) \times (H/16) \times (W/16)$ 3-D tokens and maps each token to a channel dimension of D. For trajectory input, we directly employ linear embeddings to map the channel dimension to D.

Considering the impact of the video redundancy distribution analyzed in Section III-A3 on video MDM, we design an adaptive masking strategy to standardize the modeling difficulty of videos with varying degrees of redundancy, thereby enhancing the effectiveness of pretraining. Furthermore, to jointly model the heterogeneous data of videos and trajectories, we employ MoHE Transformer as the backbone of our foundation model and design separate decoders based on vanilla Transformer for both video and trajectory inputs. Simultaneously, to model the complex associations in RS videos, we have designed an HT after the backbone to enhance the model's capability to mine intricate spatiotemporal relationships.

1) AM Strategy: Temporal redundancy in videos affects MDM, as videos have a high degree of interframe correlation, leading to substantial redundancy. Due to this interframe correlation, information from different frames can interact when modeling masked data. The interaction can result in information leakage, meaning that the information present in an unmasked frame may reveal the content that has been masked in other frames, thereby reducing the modeling complexity and impairing the effectiveness of model training. Furthermore, the presence of information redundancy lowers the modeling challenge for masked data, potentially making the model training less demanding. This, in turn, may lead to reduced robustness in real-world applications as the model fails to adapt effectively to scenarios with increased complexity. Furthermore, according to the analysis in Section III-A3, RS video data from different sources often exhibit varying levels of temporal redundancy, which further exacerbates the challenges of effectively pretraining the foundation model.

To address these issues, we propose the AM strategy, as shown in Fig. 5. First, considering the temporal redundancy in videos and the information correlation between different frames, we employ the tube masking strategy, where the same

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

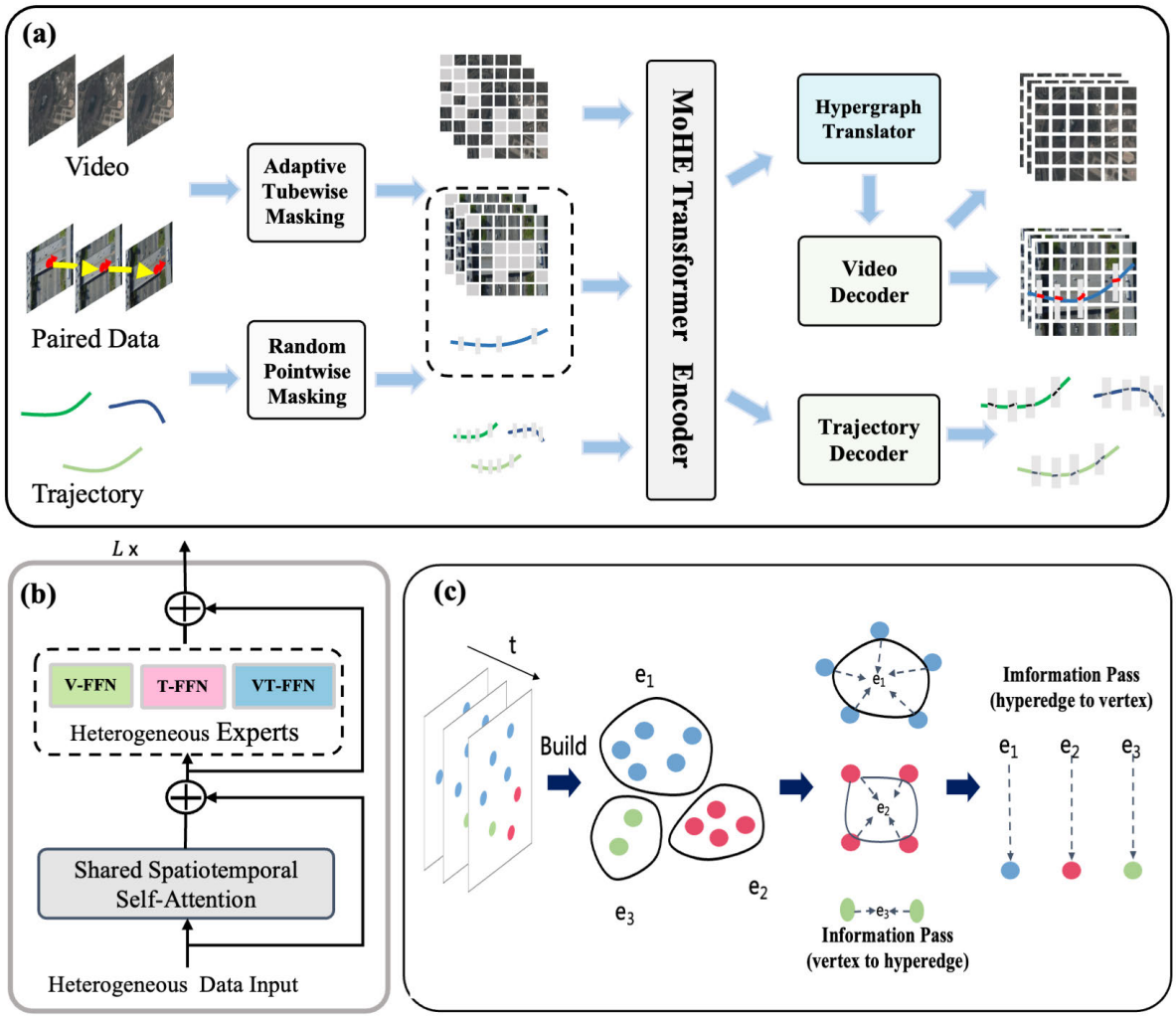


Fig. 4. (a) Overall architecture of the foundation model. (b) MoHE Transformer encoder, showing the shared spatiotemporal attention mechanism and distinct feedforward layers for different data types. (c) HT, which models intricate spatiotemporal relationships between different locations and supports complex data translations.

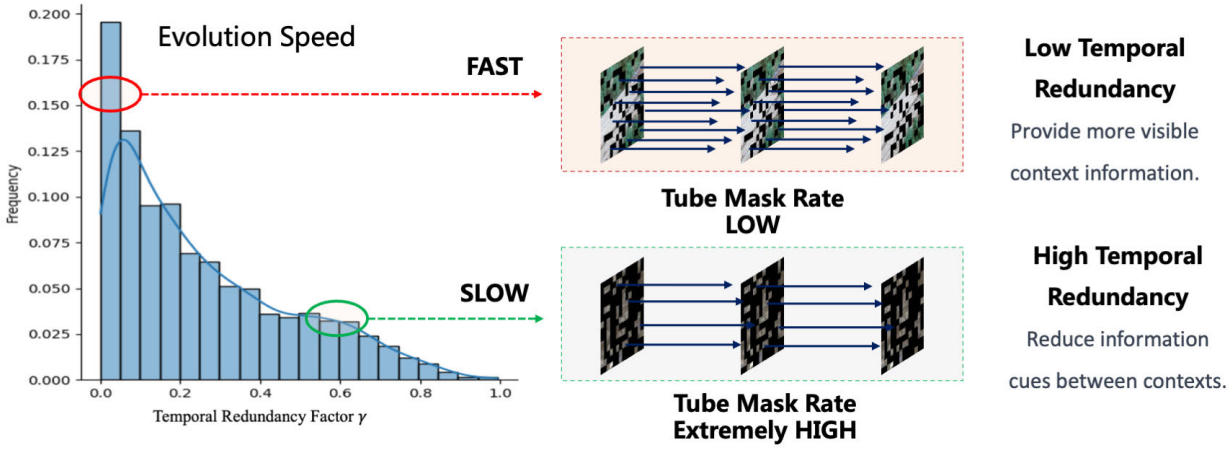


Fig. 5. Adaptively determine the mask ratio based on the object's evolution speed reflected by the video redundancy level.

positions in different frames are simultaneously masked. The temporal tube masking strategy enforces a mask to extend across the entire temporal axis, meaning that different frames share the same masking map. With the masking mechanism, the temporal neighbors of the masked cube are also masked, which implies that for slowly evolving objects, it becomes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

challenging to find their spatiotemporal correlation information across all frames, significantly reducing the risk of information leakage. Second, since RS data from different sources often exhibit varying levels of temporal redundancy, for rapidly evolving objects, solely employing the temporal tube masking strategy may still allow for the discovery of a significant amount of spatiotemporal correlation in frames that are somewhat distant. This can still result in information leakage. Therefore, we design an AM strategy, which automatically selects different tube masking ratios for various video clips. For videos with substantial temporal redundancy, we employ an extremely high masking rate to minimize the risk of information leakage as much as possible, encouraging our model to infer higher level semantics during the process of mask recovery. Conversely, for videos with lower temporal redundancy, we can achieve the same effect with a lower masking rate. Specifically, we determine the masking ratio for a video clip $\boldsymbol{X}$ based on its temporal redundancy factor $\gamma(\boldsymbol{X})$ defined in Section III-A3, which is represented by the following formula:

$$\text{ratio}(\boldsymbol{X}) = \begin{cases} 0.70, & \gamma(\boldsymbol{X}) < 0.2 \\ 0.75, & 0.2 \le \gamma(\boldsymbol{X}) < 0.4 \\ 0.80, & 0.4 \le \gamma(\boldsymbol{X}) < 0.6 \\ 0.85, & 0.6 \le \gamma(\boldsymbol{X}) < 0.8 \\ 0.90, & \gamma(\boldsymbol{X}) \ge 0.8 \end{cases} \tag{2}$$

where ratio($\boldsymbol{X}$) represents the ratio for tube masking, and the calculation method for $\gamma(\boldsymbol{X})$ is shown in (1). While alternative strategy-soft, learnable mask ratios are considered, it is not adopted for the following reason; the mask ratio, as a variable, is nondifferentiable with respect to the prediction loss. By measuring the temporal redundancy factor, we indirectly capture the evolution speed of the video sequence. A higher γ indicates lower evolution speed, and we adaptively set higher masking ratios for sequences with higher temporal redundancy to balance learning difficulty.

2) MoHE Transformer: To enable the foundation model to uniformly handle heterogeneous data, inspired by the concept of mixture of experts (MoE) networks [49], [50], [51], we introduce a universal Transformer architecture for RS spatiotemporal prediction tasks, namely, MoHE Transformer, to encode heterogeneous data input. As shown in Fig. 4(a), an MoHE layer consists of a shared spatiotemporal self-attention module and a series of feed-forward networks (FFNs). The shared spatiotemporal self-attention module can learn the alignment between video and trajectory, achieving deep integration for tasks involving heterogeneous data. The MoHE FFNs replace the original FFN, where V-FFN serves as the video expert to explore video-specific evolutionary patterns, T-FFN functions as the trajectory expert for extracting unique trajectory features, and VT-FFN acts as the video–trajectory expert, aligning features from heterogeneous data. Given the output features $\boldsymbol{H}_{l-1}$ from the previous layer, the processing of the lth layer can be represented by

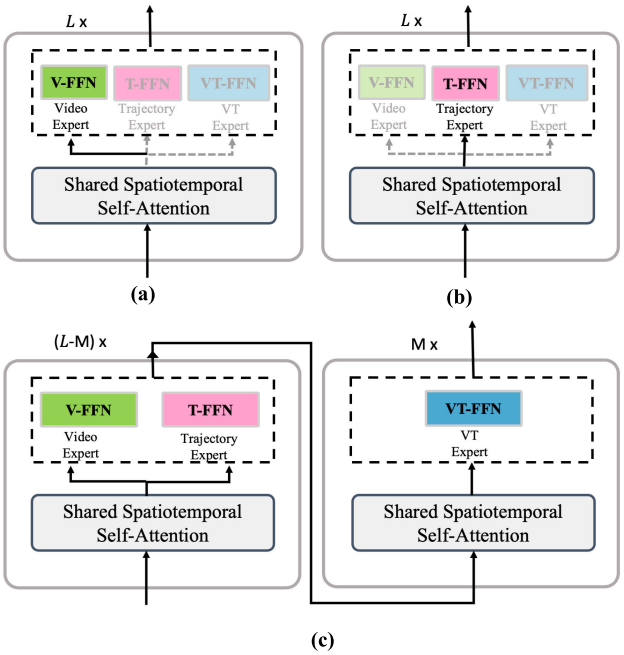


Fig. 6. Ringmo-SenseV2 can be adapted for a variety of tasks. The model automatically routes the data to the corresponding expert based on its type. (a) Video-only input. (b) Trajectory-only input. (c) Video–trajectory pair input.

the following formula:

$$\boldsymbol{H}'_l = \text{MSA}(\text{LN}(\boldsymbol{H}_{l-1})) + \boldsymbol{H}_{l-1} \tag{3}$$

$$\boldsymbol{H}_l = \text{MoHE} - \text{FFN}\big(\text{LN}\big(\boldsymbol{H}'_l\big)\big) + \boldsymbol{H}'_l \tag{4}$$

where LN stands for layer normalization and MSA represents multihead self-attention. The vectors entering the lth layer, after undergoing spatiotemporal self-attention, yield $\mathbf{H}'_l$. Subsequently, the MoHE-FFN will automatically route the data to the respective FFN expert based on the data type and the current layer l. During the pretraining phase, for video data, we pass it only through V-FFN. For trajectory data, we only make it go through T-FFN. As for video–trajectory paired data, at the lower layers of the network, we use V-FFN and T-FFN separately to learn the unique features of videos and trajectories. VT-FFN is employed at the top three layers of the network to specifically learn the alignment between video and trajectory data. In downstream tasks, different FFN experts are utilized based on the specific task and data, as illustrated in Fig. 6. When predicting spatiotemporal data for time-series grids (such as RS videos, time-series satellite images, and radar echo sequences), we can use shared spatiotemporal self-attention and V-FFN only, as shown in Fig. 4(a). When predicting various object trajectories (e.g., ships and vehicles), using shared spatiotemporal self-attention and T-FFN is sufficient, as illustrated in Fig. 4(b). To perform joint predictions on video–trajectory pairs, we employ V-FFN and T-FFN to separately learn unique features for videos and trajectories at the lower $L - M$ layers of the network. Subsequently, at the top M layer, we use VT-FFN to learn the alignment between videos and trajectories, as depicted in Fig. 4(c).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

3) Hypergraph Translator: In RS time-series data, there can be various complex dependencies among objects. For instance, the motion of one object may affect the positions of other objects or the relationships between multiple objects may change over the time sequence. This complexity goes beyond the capabilities of traditional Transformer models, as Transformers primarily focus on sequential data and may struggle to capture the nonlinear dependencies between different objects.

We introduce hypergraph to address the above issue. The hypergraph is a more flexible data structure capable of capturing more complex relationships between objects. In a hypergraph, nodes can represent objects, and edges can represent various relationships between these objects. In this article, the HT is a crucial module designed to model complex spatiotemporal dependencies among spatiotemporal locations, which cannot be effectively captured by traditional graph structures or attention mechanism. It operates in two stages: hypergraph construction and information propagation. During hypergraph construction, tokens (representing spatiotemporal features) are treated as nodes, and hyperedges are created based on token similarity, determined using a K-nearest neighbor (KNN) algorithm. This results in a hypergraph structure where each hyperedge can connect multiple nodes simultaneously, enabling the representation of higher order relationships. In the information propagation stage, the module employs hypergraph convolutions (HGNNPConv layers) to update node representations by aggregating information from hyperedges and neighboring nodes, enhancing the model's ability to capture intricate dependencies.

a) Hypergraph construction: After AM and MoHE Transformer encoder, the video input $\boldsymbol{X}$ yields tokens $V = \{v_i | i = 1, 2, \ldots, n\}$, where n is the number of tokens. After passing through the encoder, each token actually encapsulates a wealth of information, which not only encompasses the spatial and temporal features from the original video data but also captures details such as the positions, movements, and shapes of the objects within the video, and potentially their interrelationships. Consequently, these tokens transcend their role as static textual units and can be regarded as data points representing different objects and their attributes in a high-dimensional space. Hence, we can treat each token as a node and establish a hypergraph, which can model complex dependencies, including nonlinear relationships, multiobject interactions, and temporal correlations.

Specifically, the hypergraph structure is established by calculating pairwise similarity between tokens using the KNN algorithm. Each token serves as a node, while hyperedges are formed by linking each node to its K most similar neighbors. This process ensures that the hypergraph encodes meaningful spatiotemporal relationships in the data, capturing both local and global dependencies. For each data point, similarity with other data points is calculated using Euclidean distance, which can be expressed as $S_{ij} = \text{sim}(v_i, v_j)$, where v_i and v_j are feature vectors of the tokens and $\text{sim}(\cdot)$ represents the similarity measure. Subsequently, based on these similarity measures, KNNs are selected for each data point. Ultimately, through this selection of nearest neighbors, we construct a hypergraph $G = (V, E)$, where V represents the node set, E denotes the set of hyperedges, and the KNN-selected nearest neighbors form these hyperedges. Each node in V corresponds to a token, while the hyperedges in E signify the similarity relationships between the tokens.

b) Information propagation: To propagate information within the hypergraph, the module utilizes hypergraph convolution layers (HGNNPConv) [52], which aggregate features from connected nodes via hyperedges. Mathematically, the process can be described as follows:

$$\mathbf{X}' = \sigma\left(\mathbf{D}_v^{-1}\mathbf{H}\mathbf{W}_e\mathbf{D}_e^{-1}\mathbf{H}^\top\mathbf{X}\boldsymbol{\Theta}\right) \tag{5}$$

where $\mathbf{X}$ represents the input node features, $\mathbf{H}$ is the incidence matrix of the hypergraph, $\mathbf{W}_e$ denotes the edge weights, and σ represents a nonlinear activation function. This operation enables the module to capture high-order dependencies and update each node's feature representation based on its connections within the hypergraph. Unlike the way Transformers calculate attention, hypergraph convolution does not enrich a node's representation using all other nodes. Instead, it gathers the objects that are more likely to have significant associations by utilizing the hyperedges within the hypergraph, followed by information propagation and node representation updating. Additionally, batch normalization and dropout layers are applied to stabilize training and prevent overfitting.

IV. Experiment

A. Foundation Model

We validate the framework and key designs by pretraining the foundation model on the collected dataset using a random tube masking strategy with a fixed mask rate following [23]. To evaluate the impact of the HT, we incorporate it between the encoder and decoder and conduct comparative experiments under the same conditions. Additionally, to validate the effectiveness of AM strategy, we pretrain the foundation model with HT using this strategy. Downstream experiments demonstrate that Ringmo-SenseV2 can effectively model heterogeneous RS data. Importantly, it can learn the general evolution patterns of RS elements during pretraining, resulting in improved convergence speed and accuracy in downstream tasks.

All experiments with the foundation models are conducted using the PyTorch [53] deep learning framework on eight A100 GPUs. The input video has a resolution of 224 × 224 (randomly cropped from the original video with 512 × 512 resolution and resized to 224 × 224 using bilinear interpolation). The number of frames in the video and the length of trajectories are both set to 16. For the MoHE encoder, the embedding dimension D is set to 768, the number of self-attention heads is 12, and the encoder depth comprises 12 layers, with the last three layers incorporating VT-FFN to learn alignment between video and trajectory. Both the video and trajectory employ a basic Transformer decoder with a depth of six layers, and the number of self-attention heads is set to 6. We set the batch size to 480 and utilize the AdamW optimizer with a momentum of 0.9, along with a weight decay of 0.05. The optimizer hyperparameters are set as $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1\text{e}^{-8}$. The base learning rate (LR) is set to 1.5e^{-4}. We warm up the foundation model for 40 epochs

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

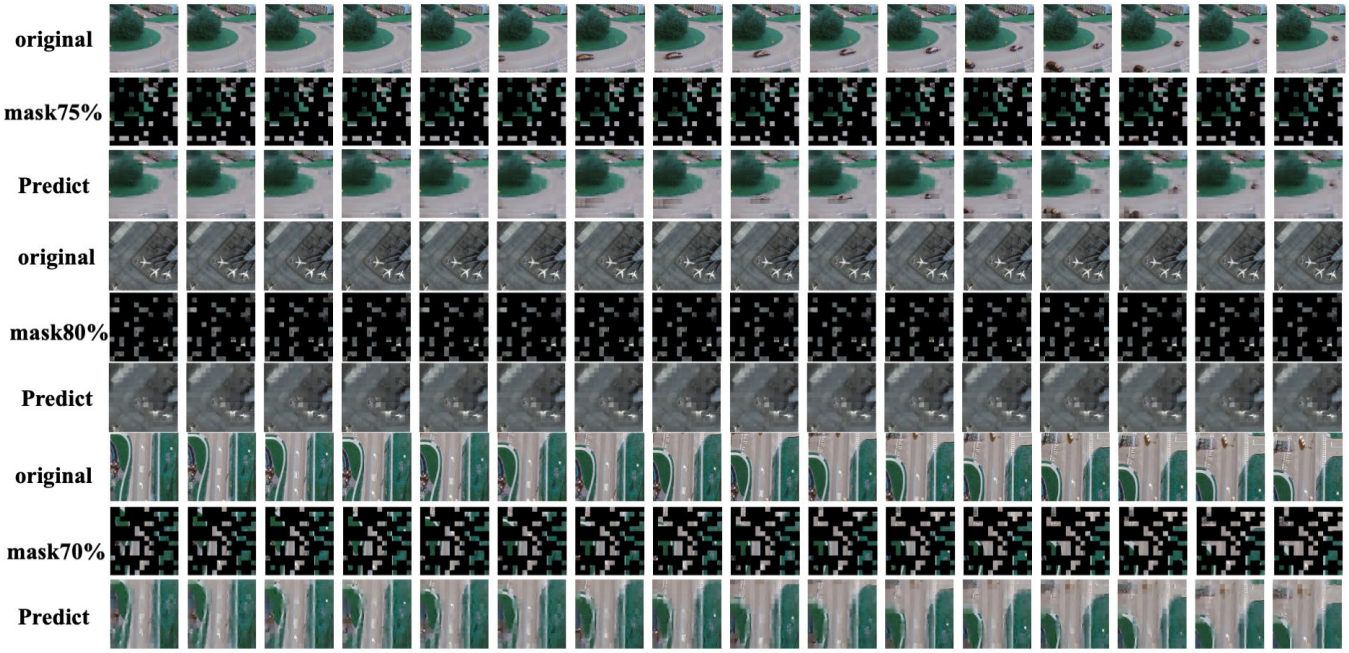


Fig. 7. Visualizations of mask and reconstruction results for the pretraining foundation model with three video samples. Due to the different temporal redundancy (evolution speed) of the three samples, the foundation model adaptively assigns different tube mask ratios to them.

and pretrain it for 300 epochs. The code and model weights will be released soon. To validate the foundation model's understanding of the spatiotemporal evolution of RS elements, we visualize the mask-prediction results of several videos and trajectories in Fig. 7. The results indicate that despite the high mask rate, our model is still capable of effectively reconstructing the masked regions of the video sequences through the unmasked contextual information and the general patterns of RS elements learned during pretraining.

After pretraining the foundation model, we integrate the pretrained backbone into downstream task models and fine-tune accordingly. We validate the effectiveness of the foundation model across various common RS prediction tasks, including RS video prediction, Earth surface prediction, RS segmentation forecasting, ship trajectory prediction, and RS video–trajectory joint prediction. Furthermore, in addition to these prediction tasks, we extend Ringmo-SenseV2 to include the scene graph generation task for RS images, assessing its generalization performance in image-related tasks.

B. Earth Surface Forecasting

Earth surface forecasting is a significant research area aimed at predicting the normalized difference vegetation index (NDVI) of the Earth's surface over a certain period in the future. This prediction is based on historical satellite time-series images and future weather conditions. NDVI is a crucial ecological indicator commonly used to measure the extent of green vegetation cover on the Earth's surface, reflecting the health and changes within ecosystems. The objective of Earth surface forecasting is to provide accurate forecasts of future NDVI values by combining historical satellite imagery and future meteorological data. This task holds promising applications in various fields, including but not limited to ecological monitoring, agricultural production, water resource management, drought monitoring, and environmental conservation. In the experiments for Earth surface forecasting, we utilize an adapted version of SimVP implemented in https://github.com/earthnet2021/earthnet-models-pytorchthe official codebase of EarthNet. We replace SimVP's translator with the MoHE backbone while keeping the other structures unchanged.

1) Dataset Description: We utilize the EarthNet2021x dataset to validate the performance of the foundation model in Earth surface vegetation prediction [44]. EarthNet2021 is a large-scale dataset and challenge focused on Earth surface prediction, defining Earth surface forecasting as the task of predicting satellite images under the constraint of future weather conditions. The EarthNet2021x dataset is reprocessed data from the EarthNet2021 challenge, with a specific focus on vegetation forecasting in the European region. The data are organized into minicubes as the basic units, with each minicube containing 30 frames of high-resolution (20 m) satellite imagery in four channels (blue, green, red, and near infrared) captured every 5 days, 150 frames of five meteorological variables (captured daily), and a static digital elevation model at medium to high resolution. We train the model using all the samples from the training set and evaluate the model's performance on the IID test set, according to the split of the official research. The NDVI can be calculated using the following formula:

$$\mathrm{NDVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}} \tag{6}$$

where NDVI is the normalized difference vegetation index, NIR is the reflectance in the near-infrared spectrum, and Red is the reflectance in the red spectrum.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

2) Implementation Details: The experiments are conducted using the PyTorch deep learning framework [53] on two A40 GPUs. We set the batch size to 8 and utilize the first 10 frames from minicube along with future meteorological variables to jointly predict the NDVI values for the next 20 frames. We use the AdamW optimizer to minimize the $L2$ loss for NDVI values, with optimizer parameters set to $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1 \times 10^{-6}$. The number of training epochs is set to 100, and the multistep LR scheduler is used during training. The depth of both the encoder and decoder in the SimVP architecture is set to four layers, and the channel dimension is set to 128. We use vegetation score, RMSE, and R2 as evaluation metrics to measure the quality of model predictions. Vegetation score, defined in the EarthNet2021 challenge, is calculated as follows:

$$\mathrm{NSE} = 1 - \frac{\sum_{i=1}^{n}(T_i - P_i)^2}{\sum_{i=1}^{n}(T_i - \overline{T})^2} \tag{7}$$

$$\mathrm{Score}_{\mathrm{veg}} = \frac{1}{2 - \mathrm{NSE}} \tag{8}$$

where nash–sutcliffe model efficiency (NSE) measures the goodness of fit between the model's predicted values (P_i) and the actual values (T_i), and $\overline{T}$ represents the mean of the actual values. A higher NSE value closer to 1 indicates a better fit of the model, while an NSE value closer to 0 or negative implies a poorer predictive performance of the model. The range of NSE extends from negative infinity to 1, while, through (8), the range of $\mathrm{Score}_{\mathrm{veg}}$ is mapped to the interval from 0 to 1.

3) Comparison With the SOTA Methods: To validate the performance of the pretrained foundation model, we compared it under the same experimental conditions with contemporary classical spatiotemporal prediction methods, including SimVP [54], PredRNN [11], and Channel-U-Net [44]. SimVP is a simple yet effective video prediction model built entirely on convolutional neural networks (CNNs) and trained end-to-end with mean squared error (mse) loss. PredRNN employs a pair of explicitly decoupled memory units that deal with spatial and temporal dynamics at different hierarchical levels, ultimately forming a unified representation of complex environments. It has shown highly competitive results in predictive learning scenarios under both no-motion and motion conditions. Channel-U-Net is a baseline network proposed by EarthNet2021, utilizing the U-Net architecture with dense connections between layers. All available context time step inputs are stacked along the channel dimension and fed into the network. Subsequently, the model outputs all future time steps, which are also stacked along the channel dimension and then reshaped for proper evaluation. The experimental results are shown in Table II.

Classical video prediction methods show subpar performance on EarthNet2021x, especially PredRNN, which achieves a vegetation score of only 0.272. SimVP, as an outstanding work that employs CNNs for video prediction, demonstrates slightly better performance with a vegetation score of 0.282. However, Channel-U-Net outperforms the rest, achieving a vegetation score of 0.311. Our method, leveraging the powerful spatiotemporal feature learning capability of Transformers, combined with the vast dataset of EarthNet2021x and the prior knowledge of spatiotemporal evolution of RS elements within the foundation model, ultimately achieved the highest vegetation score of 0.415.

TABLE II

COMPARISON OF EARTH SURFACE FORECASTING EXPERIMENTAL RESULTS. BOLD INDICATES THE BEST AND UNDERLINE INDICATES THE SECOND BEST. "W/O" STANDS FOR "WITHOUT," "AM" STANDS FOR "ADAPTIVE TUBE MASKING STRATEGY," AND "HG" STANDS FOR "HYPERGRAPH TRANSLATOR"

Models	VegS↑	RMSE↓	R2↑
SimVP[55]	0.282	0.149	0.495
PredRNN[13]	0.272	0.151	0.496
Channel-U-Net[45]	0.311	0.153	0.526
Ringmo-SenseV2	**0.415**	**0.136**	**0.555**
w/o AM	0.411	**0.136**	0.550
w/o AM,HG	0.400	0.138	0.544
Train from scratch	0.393	0.139	0.541

Furthermore, when we sequentially load the pretrained parameters of the foundation model without using the AM strategy and without using this strategy along with the HT, the model's performance exhibits a decrease. When the pretrained parameters of the foundation model are not loaded, the vegetation score of the model decreases further. This confirms the effectiveness of AM and HG.

4) Visualizations: Fig. 8 displays visualization results of various methods. We use different shades of green to represent NDVI values, where darker colors indicate NDVI values closer to 1, indicating healthier surface vegetation. While classical methods provide an overall prediction of vegetation distribution, they fail to predict changes in small local areas with detail. To intuitively demonstrate the prediction performance of each method, we have outlined a local area in the last frames using a slanted rectangular box. In the ground truth, this area appears relatively light, suggesting either the absence of vegetation or sparse vegetation cover. However, in the predictions from SimVP, PredRNN, and Channel-U-Net, this area appears very dark, which is attributed to the blurriness caused by long-term forecasting. SimVP (replacing translator), which was trained from scratch without loading pretrained parameters, exhibits slightly improved performance. In stark contrast to the classical methods, our model predicts this area with light colors, closely resembling the ground truth, highlighting the advantages of the foundation model.

C. RS Video Prediction

RS video prediction, which involves forecasting future frames in videos captured by RS technologies such as satellites and drones, is an important research field with vast practical applications. This task includes accurate forecasting of various scenarios, environments, and natural phenomena on the Earth's surface. Such predictions have wide-ranging applications in fields such as meteorology, climatology, geology, natural disasters, agriculture, and urban planning. In the experiments for RS video prediction, we replace the translator of SimVP [54] with the MoHE backbone while keeping the rest of the structure identical to it.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

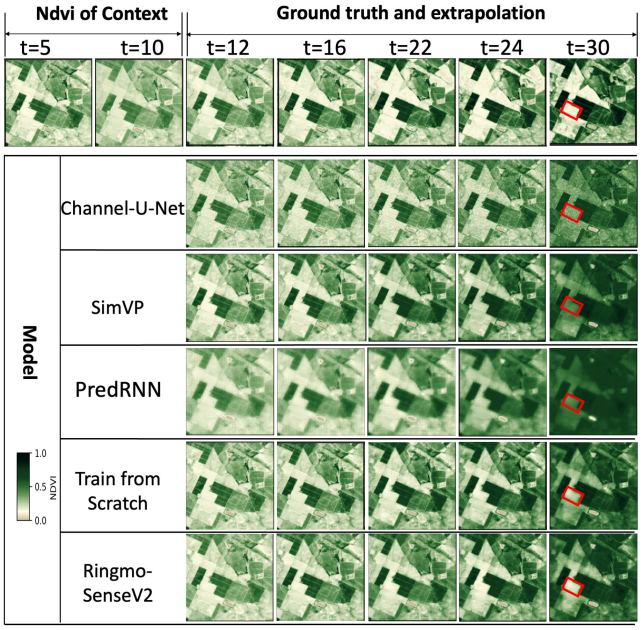


Fig. 8. Visualization results of Earth surface forecasting comparative experiments.

1) Dataset Description: As there is no available video prediction dataset in RS, we transform the Event Recognition in Aerial videos (ERA) dataset [55], which is originally designed for RS video understanding, into a video prediction dataset. ERA is a large and diverse dataset that encompasses various environmental scenarios and shooting scales, designed to train models for understanding events in UAV videos. It comprises 2864 videos, covering 25 different event classes. Each video sequence is recorded at 24 frames/s, has a duration of 5 s, and features a spatial resolution of 640 × 640 pixels. We perform nonoverlapping sampling using a sliding window of length 40 in each video. Subsequently, we select 8 frames at regular intervals within each window to create a video prediction sample, with the first 4 frames used as input and the following 4 frames as labels. The spatial resolution is resized to 256 × 256 using bilinear interpolation. In the end, we obtain 4415 video sequences for training, 1250 video sequences for validation, and 2918 sequences for testing.

2) Implementation Details: All experiments are conducted using the PyTorch deep learning framework on a single A40 GPU [53]. The depth of both the encoder and decoder in the SimVP architecture is set to four layers, and the channel dimension is set to 128. We use the Adam optimizer with a momentum of 0.9 and employ the cosine LR scheduler to fine-tune the model with batch size 2 for 50 epochs. The minimum LR is set to $1e^{-6}$, and the LR is set to $1e^{-3}$. For all experiments, we fine-tune all parameters simultaneously. We use four metrics, namely, mse, mean absolute error (MAE), SSIM, and peak signal-to-noise ratio (PSNR), to assess the models' performance.

3) Comparison With the SOTA Methods: In order to demonstrate the superiority of our foundation model, we compared it with classical video prediction models, including ConvLSTM [9], masked image modeling (MIM) [56], PhyDNet [57], and SimVP [54]. ConvLSTM employs convolutional structures in both the input-to-state and state-to-state transitions of LSTM to capture spatial features and temporal dependencies, making it one of the early attempts at video prediction using deep learning methods. MIM models spatiotemporal dynamics more effectively by explicitly handling stationary and nonstationary components. PhyDNet is a deep model that utilizes physical knowledge to guide video prediction, which introduces a novel physics unit called PhyCell, inspired by data assimilation techniques, to predict PDE-constrained outcomes in latent space. SimVP is a simple yet effective video prediction model built entirely on CNNs and trained end-to-end with mse loss. SimVP achieves SOTA performance on multiple benchmark datasets without the need for complex modules, strategies, or tricks. It also demonstrates strong generalization and scalability, even on real-world datasets. All the code implementations followed the OpenSTL code repository, and the optimizer, LR, and batch size are kept consistent with what was described in Section IV-C2. The experimental results are shown in Table III. If training SimVP (with the translator replaced) from scratch, the model achieves an SSIM score of only 0.566 on the ERA dataset, slightly lower than the original SimVP. However, when using our pretrained parameters, the SSIM score increases to 0.573, and the PSNR reaches a maximum of 21.94. Loading the foundation model parameters pretrained without the AM strategy leads to a slight decrease in the final results. When loading pretrained parameters without the adaptive mask strategy and without adding the HT, the model's performance decreases again, demonstrating the effectiveness of our proposed strategy and module.

TABLE III
COMPARISON OF VIDEO PREDICTION TASK EXPERIMENTAL RESULTS. BOLD INDICATES THE BEST AND UNDERLINE INDICATES THE SECOND BEST. "W/O" STANDS FOR "WITHOUT," "AM" STANDS FOR "ADAPTIVE TUBE MASKING STRATEGY," AND "HG" STANDS FOR "HYPERGRAPH TRANSLATOR"

Models	MSE↓	MAE↓	SSIM↑	PSNR↑
ConvLSTM[9]	2366.14	13506.07	0.488	20.77
MIM[57]	2102.70	12144.84	0.536	21.57
PhyDNet[58]	2566.30	14734.49	0.483	20.18
SimVP[55]	2049.15	11763.86	0.568	21.88
Ringmo-SenseV2	2028.86	11742.59	**0.573**	**21.94**
w/o AM	2029.68	**11691.24**	0.572	21.93
w/o AM,HG	**2019.61**	11775.07	0.570	21.91
Train From Scratch	2030.31	11929.63	0.566	21.84

4) Performance for Long-Term Prediction: To evaluate the model's performance for long-term prediction, we conduct additional experiments on the RS video prediction task. Beyond the original setup, where the model predicts the next 4 frames based on the initial 4 frames, we extend the testing to predict 8, 12, 16, 20, and 24 subsequent frames. Specifically, the first 4 frames are used to predict frames 5–8, the predicted frames 5–8 are then used as input to forecast frames 9–12, and this process is repeated iteratively for longer predictions. As shown in Table IV, the results indicate that while the model's accuracy decreases as the prediction horizon extends, the decline is slower than linear. This demonstrates the model's

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
MODEL'S PERFORMANCE AS THE NUMBER OF PREDICTED FRAMES INCREASES

Configs	MSE↓	MAE↓	SSIM↑	PSNR↑
4 → 4	2028.86	11742.59	0.573	21.94
4 → 8	2682.57	13848.92	0.530	20.76
4 → 12	3242.85	15661.67	0.499	19.86
4 → 16	3716.31	17144.82	0.479	19.19
4 → 20	4139.64	18410.51	0.464	18.67
4 → 24	4510.12	19498.77	0.453	18.24

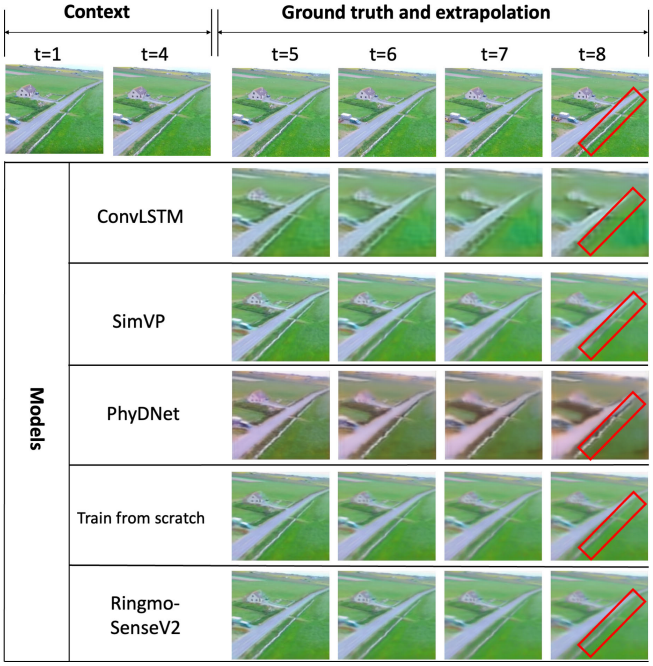


Fig. 9. Visual results of the RS video prediction comparative experiments.

robustness and its potential for long-term forecasting. The findings also provide valuable insights into mitigating error accumulation, offering directions for further optimization in future work.

5) Visualizations: Fig. 9 displays the prediction results of different models for a specific test sample. We select a sequence captured by a drone, showing a car exiting a village intersection, as the test sample. As for the clarity of the prediction results, it is evident that as the prediction time extends, the predicted frames from each model gradually become blurry. However, overall, the model loaded with our pretrained parameters exhibits relatively lower blurriness.

For a more intuitive comparison of the prediction results, we select the same region for comparison among the last frames of ground truth and prediction results, indicated by slanted rectangular boxes. In the ground truth, there is a road passing through the center of the box, clearly visible. In the prediction result of ConvLSTM, this road is indistinguishable due to blurriness. In the SimVP prediction result, the road can be vaguely seen but is also quite blurry. In the PhyDNet prediction result, the road is visible but deformed and noticeably shifted upward. For our SimVP (with translator replaced) prediction results, the road is relatively blurry and shifted upward, but after loading the pretrained parameters, the road's position significantly corrects, and the clarity slightly improves. This demonstrates that the base model helps improve both the clarity and the accuracy of the object's position in video prediction.

D. RS Segmentation Forecasting

There are some works on generating future images of an interesting region and evaluating the valuable indexes of the area. However, we hold a novel point that sometimes the intermediate pixel values of the image bands are not necessary, we care more about the labels of the region of interests (ROIs).. Thus, we propose to forecast the future labels directly from the previous time-series images, which we call RS segmentation forecasting.

1) Dataset Introduction: As there are no available datasets for this novel task in the field of RS, we transform two commonly used time-series datasets with segmentation labels to the forecasting format and conduct experiments on the datasets. One dataset we used is UAVid [58], which is a UAV video dataset with a total of 42 video sequences (20 for training, 7 for validation, and 15 for testing). The spatial resolution of the images is 4k. There are eight labeled classes in UAVid. The other dataset we used is SpaceNet7 [59], which contains 55 satellite image sequences with two labeled classes, e.g., building and background. We use 50 for training and 5 for validation. The spatial resolution of the images is 1024×1024.

As for the dataset format, we follow the data organization of UAVid broadly and transform these datasets to the same format. However, we select several historical images and the true future label as input instead of the image–label pair as the normal segmentation task.

2) Implementation Details: To pursue a fair comparison with other methods, we build a unified framework for RS segmentation forecasting. We find that many SOTA methods in this research field are not open source, so we reimplement them according to their papers and report the reimplementation results here. For compatibility with upstream pretraining, we select UNetformer [60] as the basic model and expand it along the temporal dimension. The implementation details are as follows.

The designed model consists of three main components: the spatial information extractor, the temporal correlation miner, and a sophisticated segmentation head. To get the performance upper bound of the forecasting model, we replace the backbone of UNetformer with the plain ViT and train the model following the standard image segmentation pipeline. We make this decision according to the structure of the upstream model. We also build a spatiotemporal version of UNetformer with the SimVP module as the temporal correlation miner.

We crop the spatial resolution of the time-series clip into 512×512, as for the temporal dimension, we input four historical images. We train our model for 40 000 iterations on both datasets. The LR is set to 0.0002 at the beginning and decays to 0.0 with the poly strategy.

3) Comparison With SOTA Methods: We follow previous works and set the plain UNetformer model trained on image

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

TABLE V
COMPARISON OF RS SEGMENTATION FORECASTING TASK EXPERIMENTAL RESULTS ON MIOU (%)

Method	UAVid	SpaceNet7
UNetformer(Oracle)	74.15	64.18
Copy Last	61.98	63.24
StF (UNet)	62.39	60.96
UNetFormer[61]	64.58	63.55
Ringmo-SenseV2	**65.08**	**63.9**

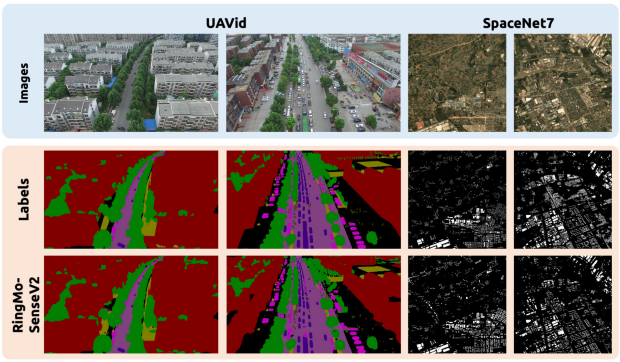


Fig. 10. Visualization of RS segmentation forecasting. We visualize the ground truth and our Ringmo-SenseV2 results and observe that Ringmo-SenseV2 can produce satisfactory forecasting results for most scene elements.

datasets as the upper bound of the forecasting performance. We also use the "copy last" results as the lower bound. As there are no available codes for the latest SOTA methods, we report our reimplemented results for these methods here.

It is worth noting that UNetformer (Oracle) is set as the upper bound of the forecasting performance, while "Copy Last," which copies the segmentation results of the last input frame as the forecasting output, is set as the lower bound. According to the results in Table V, we observe that RingMo-SenseV2 indeed enhances the extracted spatiotemporal feature and boosts the performance of RS segmentation forecasting. Also, we find that RingMo-SenseV2 performs much better on both UAVid and SpaceNet7. We think that the reason is that RingMo-SenseV2 is pretrained on the collected dataset that contains RS time-series images, making it more tailored to RS segmentation datasets.

E. Visualization

We visualize the RingMo-SenseV2 results to show the effectiveness of our method more visually. As shown in Fig. 10, the RingMo-SenseV2 can forecast the grand scenes well, e.g., the road and building in UAVid. When it comes to small objects and moving objects(e.g., humans and moving cars in UAVid), the forecasting results see some degradation. We think that the reason is that the small objects and moving objects are not stable in the temporal dimension, which is hard to be captured. In conclusion, our RingMo-SenseV2 has captured the macro spatiotemporal information of the RS images, which is beneficial to the RS segmentation forecasting task, while the small objects and moving objects are still challenging for the forecasting task.

F. Ship Trajectory Prediction

Ship interaction systems are crucial in many applications, especially in autonomous navigation. Accurately predicting the future paths of ships is vital. Ship trajectory prediction is complex due to the uncertainties in each ship's future movements and the unclear intentions behind their actions. We view ship trajectory prediction as a task of modeling the future paths of N ships based on their past paths.

1) Dataset Description: The model underwent meticulous fine-tuning for the trajectory prediction task, utilizing the ship trajectory dataset featuring a diverse array of 19 915 vessels. To imbue the trajectories with a sense of realism, preprocessing measures were judiciously applied. Anomalously high- and low-speed ship trajectories, defined by thresholds of 0 and 120 km/h, were systematically removed. Subsequently, the trajectory data underwent bilinear interpolation, ensuring a uniform 15-min interval sampling and encompassing 28 coordinate points per trajectory. The resultant preprocessed dataset comprises a substantial total of 37 079 trajectories, with 70% allocated to the training set and the remaining 30% constituting the test set. This methodical partitioning strategy safeguards against biases, providing a robust foundation for model evaluation and validation.

2) Implementation Details: We extrapolate the trajectories for the subsequent 12 time steps based on the observed trajectories spanning eight time steps, concurrently enhancing the training data with randomly rotated scenes. Our prediction model is anchored in the utilization of two identical basic Transformer layers in each encoder/decoder, characterized by a dropout rate of 0.1. The model's architecture mandates the dimensions of query, key, and time coding to be uniformly set at 504, with the hidden dimensions of the feedforward layer resonating at 2048. The multihead attention is executed with a complement of 12 heads, and the hidden dimensions across all multilayer perceptrons (MLPs) within the model stand at 2048. The training process is guided by the mse as the loss function, spanning a rigorous 100 epochs facilitated by the Adam optimizer.

The LR initiates at 0.0001, systematically halving every ten epochs to optimize model convergence. Evaluation metrics hinge on ADE_K/FDE_K, where ADE_K is defined as $\mathrm{ADE}_K = (1/T)\min_{k=1}^{K}\sum_{t=1}^{T}\|\hat{\mathbf{y}}_n^{t,(k)} - \mathbf{y}_n^t\|$, and FDE_K is defined as $\mathrm{FDE}_K = \min_{k=1}^{K}\|\hat{\mathbf{y}}_n^{T,(k)} - \mathbf{y}_n^T\|$, where $\hat{\mathbf{y}}_n^{T,(k)}$ is the coordinate of agent n at future time t in the kth sample, $\mathbf{y}_n^T$ denotes the corresponding ground truth, and $\|*\|$ denotes the two-norm of $*$. To introduce multimodal trajectories, K independent and uniformly distributed noises, within the interval $[-0.04, 0.04]$, are injected into the single trajectory output from the model. Subsequent metric computations are based on a robust sampling number of $K = 20$, offering a comprehensive evaluation of the model's predictive versatility.

3) Comparison With the SOTA Methods: We have selected a diverse array of methods as baselines for our evaluation, encompassing traditional trajectory prediction

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPARISON OF EXPERIMENTAL RESULTS FOR THE SHIP TRAJECTORY PREDICTION TASK

Method	$ADE_{20}\downarrow$	$FDE_{20}\downarrow$
Kalman Filtering[62]	0.1764	0.3272
RNN[63]	0.1104	0.1940
LSTM[64]	0.1092	0.1839
Transformer-TF[65]	0.2227	0.5111
AgentFormer[66]	0.1430	0.2214
Ringmo-SenseV2	**0.0885**	**0.1349**

techniques, classical trajectory prediction methodologies, and the most proficient trajectory prediction models. The chosen methods include Kalman filtering [61], RNN [62], LSTM [63], Transformer-TF [64], and AgentFormer [65]. Table VI meticulously delineates the accuracies attained by each method in the ship trajectory prediction task. Kalman filtering achieves ADE_{20}/FDE_{20} values of 0.1764/0.3272 on the ship trajectory dataset, while RNN and LSTM exhibit improved performance with ADE_{20}/FDE_{20} values of 0.1104/0.1940 and 0.1092/0.1839, respectively. Transformer-TF registers ADE_{20}/FDE_{20} values of 0.2227/0.5111, and AgentFormer attains ADE_{20}/FDE_{20} values of 0.1430/0.2214 on the ship trajectory dataset. In contrast, our model, enriched with RingMo-SenseV2 pretraining parameters, showcases a noteworthy reduction in ADE_{20}/FDE_{20} to 0.0885/0.1349. This outcome underscores the efficacy of RingMo-SenseV2 in adeptly capturing spatiotemporal interaction features among agents in the ship trajectory prediction task, leading to a discernible decrease in trajectory prediction errors. The utilization of RingMo-SenseV2 pretraining parameters thus accentuates the model's capacity to extract nuanced features and enhance predictive accuracy in complex multiagent scenarios.

4) Visualizations: Fig. 11 provides a visual representation of the outcomes from the ship trajectory prediction task, illuminating the inherent uncertainty embedded within each agent's future behavioral trajectory. This complexity presents a formidable challenge to the model's capacity to discern and encapsulate the nuanced navigational intent inherent in each ship's trajectory. The predicted trajectories presented by the baseline models within the figure conspicuously deviate from the actual values, and conversely, the trajectory predictions emanating from RingMo-SenseV2 exhibit a notable enhancement in accuracy. The discernibly superior alignment between the predicted trajectories and the actual values attests to the efficacy of RingMo-SenseV2 in surmounting the challenges posed by complex agent interactions and uncertainties in individual ship trajectories. This visual evidence serves to underscore RingMo-SenseV2's prowess in capturing the intricacies of navigational intent, positioning it as a commendable solution that outperforms conventional baselines in the realm of trajectory prediction.

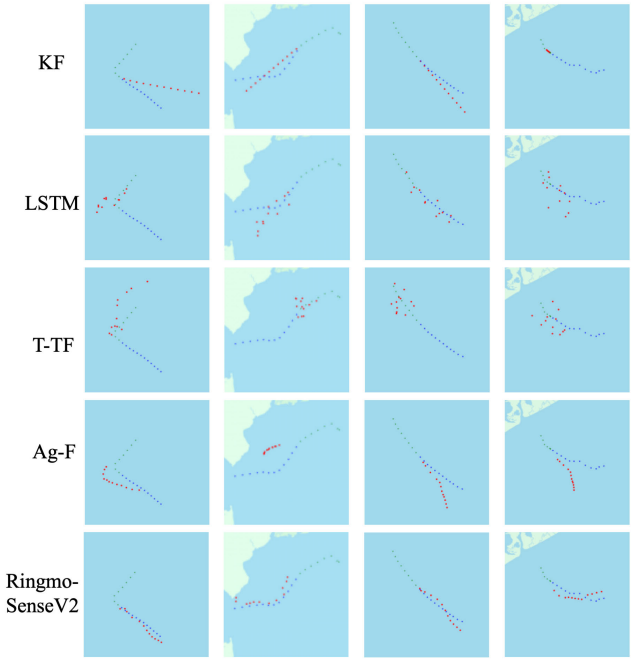


Fig. 11. Visual results of the ship trajectory prediction experiments. T-TF denotes Transformer-TF and Ag-F denotes AgentFormer.

G. RS Video–Trajectory Joint Prediction

The RS video–trajectory joint prediction task is a novel and forward-looking research challenge. Its objective is to simultaneously predict future video frames and the future positions of objects of interest by combining historical video frames and the corresponding object location information. Unlike traditional video prediction tasks, this task emphasizes innovation and uniqueness in the RS domain. It goes beyond predicting only future video frames and provides specific information about the positions of objects of interest, which is a crucial aspect not previously explored in past research. In RS video–trajectory joint prediction, we break free from the constraints of predicting solely future video frames. Instead, we focus on the trajectories and location information of objects of interest, combining them with predictions of future video frame content. This innovative task holds vast potential for various applications, including but not limited to traffic management, urban planning, disaster monitoring, and environmental protection. Through this task, we have the opportunity to better understand the spatiotemporal relationships in the RS field, providing richer information for practical applications. To our knowledge, we are the first team to propose and study this innovative task. This opens up new directions and opportunities for future research in the RS field, broadening our understanding and applications of spatiotemporal information. In the experiments for RS video–trajectory joint prediction, we replaced SimVP's translator with the MoHE backbone, as shown in Fig. 6(c). The structural aspects related to video prediction are kept consistent with SimVP. Additionally, we extend the backbone's output with a straightforward Transformer-based trajectory prediction head for handling trajectory features [66].

1) Dataset Description: To the best of our knowledge, the task of video–trajectory joint prediction has not been previously studied, and there is no dedicated dataset for experiments in this task. Therefore, we consider adapting a tracking dataset

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

for video–trajectory joint prediction. For this purpose, we use the UAV123 dataset [67]. UAV123 is a dataset for long-term aerial tracking. It consists of a total of 123 video sequences with over 110 000 frames, and all sequences are fully annotated with upright bounding boxes, making it possible for us to transform it into a dataset for video–trajectory joint prediction. We use a sliding window of 40 frames with a step size of 3 for sampling. After obtaining several 40-frame samples, we further sample 16 frames (8 frames for input and 8 for labels) with a 2-frame interval starting at any possible location to get the final samples. The spatial resolution of all video frames is scaled to 256×256 using bilinear interpolation. The trajectory points of the objects of interest are represented by the central pixel coordinates of their annotated bounding boxes. Finally, we obtain a total of 10 172 samples, with 7106 used for training, 2043 for validation, and 1023 for testing.

2) Implementation Details: All experiments use the PyTorch deep learning framework and are conducted on a single A40 GPU. We set the batch size to 2 and use the image sequences of the first 8 time steps along with the corresponding trajectories to predict the image sequences and trajectories for the subsequent eight time steps. We fine-tune the model using the Adam optimizer with a momentum of 0.9, weight decay set to 0, and a cosine LR scheduler. The minimum LR is set to $1e^{-6}$, and the initial LR is set to $1e^{-3}$. In the SimVP architecture, both the encoder and decoder have a depth of four layers, and the channel dimension is set to 128. For the trajectory decoder, the depth is set to six layers, and the embedding dimension is set to 512, consistent with the backbone. We compute the video output loss using mse and the trajectory loss using the Euclidean distance between predicted trajectory points and ground truth trajectory points. The final loss calculation formula is as follows:

$$L = \frac{1}{T}\sum_{t=1}^{T}\left\|I_t^{\text{true}} - I_t^{\text{pred}}\right\|_2^2 + \lambda \cdot \left\|P_t^{\text{true}} - P_t^{\text{pred}}\right\|_2 \quad (9)$$

where λ controls the balance between the two losses, and in our experiments, we set it to 0.1. For video output, we use SSIM and PSNR as evaluation metrics, consistent with video prediction tasks. For trajectory output, we use FDE and ADE as evaluation metrics.

3) Comparison With the SOTA Methods: To validate the effectiveness of the foundation model and the video–trajectory joint prediction task, we compare our method under the same experimental conditions with classical video prediction methods PhyDNet [57] and PredRNN [13], both of which were introduced earlier. The results are shown in Table VII. From various evaluation metrics, it is evident that our model, with the foundation model pretraining parameters loaded, outperforms traditional computer vision spatiotemporal prediction methods. Furthermore, when loading the foundation model parameters without using the AM strategy or without pretraining with this strategy and without the HT, the model's performance decreases. When the foundation model parameters are not loaded, and training starts from scratch, the performance further decreases, which further demonstrates the effectiveness of the designed AM strategy and HT.

TABLE VII
COMPARISON EXPERIMENTS FOR VIDEO–TRAJECTORY JOINT PREDICTION. BOLD INDICATES THE BEST AND UNDERLINE INDICATES THE SECOND BEST

Models	SSIM↑	PSNR↑	FDE↓	ADE↓
PhyDNet[58]	0.448	20.09	——	——
PredRNN[13]	0.469	21.31	——	——
Ringmo-SenseV2	**0.477**	**21.31**	**5.48**	**4.08**

4) Visualizations: For a more intuitive presentation of the prediction results of each model, we provide visualizations of the main experimental results in Figs. 12 and 13. Fig. 12 displays a video sample captured by a drone, showing vehicles turning and diverting at a roundabout. The small red dots in the image indicate the tracked vehicles of interest. From the experimental results, it is evident that PhyDNet's predictions start to blur as early as the tenth frame, becoming nearly unrecognizable in terms of the road shape by the final frame. PredRNN performs slightly better. In comparison, the area we circled with an ellipse in the last frame of our model's prediction is relatively clear, although it becomes somewhat blurrier when training from scratch without loading pretrained parameters. Fig. 13 illustrates the trajectory prediction results under two experimental conditions: one with pretrained foundation model parameters loaded and the other starting training from scratch without loading parameters. In this figure, blue line segments represent the trajectories for the first eight time steps, green line segments represent the ground truth for the following eight time steps, red denotes the predictions from the model trained from scratch, and yellow represents the predictions from the model with pretrained parameters loaded. While the overall trend of the model trained from scratch is similar to the ground truth, individual trajectory points are noticeably shifted upward. In contrast, the model with pretrained parameters produces predictions where individual trajectory points closely align with the ground truth, highlighting the foundation model's generalization ability and guiding role in trajectory prediction.

H. RS Scene Graph Generation

The RS scene graph generation task [68] focuses on constructing scene graphs from RS images by identifying objects and their relationships, thereby enhancing graph reasoning for advanced tasks such as visual captioning and visual question answering (VQA). RS images present unique challenges due to their high-altitude perspective, complex multiscale features, and diverse relationships among objects. Our proposed HT, adept at managing complex spatiotemporal dependencies, may help address these challenges. Although Ringmo-SenseV2 was initially pretrained on time-series data, its flexible Transformer architecture allows for straightforward adaptation to image tasks. To assess the generalizability of Ringmo-SenseV2 in generating scene graphs, we test its performance on this task.

1) Dataset Description: To construct an RS scene graph generation dataset equipped with oriented bounding box annotations, we utilize FAIR1M [69] as the foundation for our

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

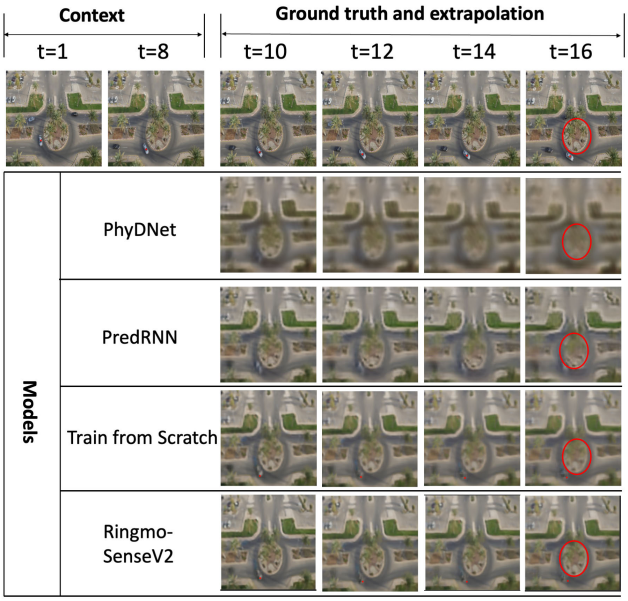


Fig. 12. Visualization results of video–trajectory joint prediction comparative experiments.

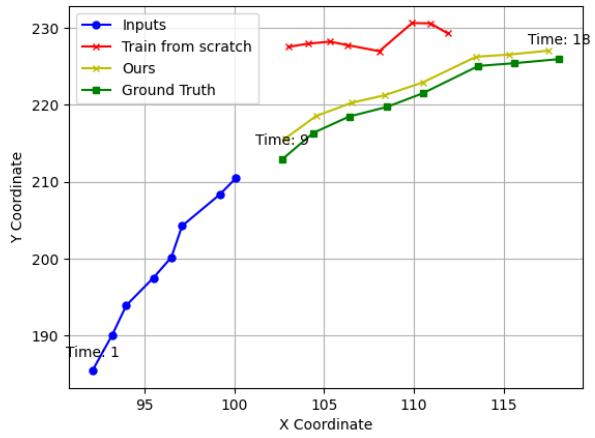


Fig. 13. Comparison of prediction performance with and without loading the foundation model parameters for a sample, where the trajectory is from the vehicle making a turn on the roundabout in Fig. 12.

dataset construction. FAIR1M serves as a large-scale benchmark dataset for fine-grained object detection and recognition tasks within high-resolution RS imagery. It comprises over 1 million instances and more than 40 000 images, collected from high-resolution satellites and Google Earth, with resolutions ranging from 0.3 to 0.8 m. FAIR1M employs oriented bounding box annotations to label objects, categorizing them into 37 fine-grained categories. We directly utilize FAIR1M's imagery as our source, excluding a minor fraction of images that do not exhibit clear relationships. Furthermore, the dataset division allocates half of the images to the training set, (1/6) to the validation set, and (1/3) to the test set. The dataset includes original images along with their corresponding oriented bounding box annotations and relationship labels.

The original FAIR1M dataset annotations include only five categories, which are inadequate for our needs. To address this, we preserved the original annotations but expanded the dataset to include a total of 60 object categories and introduced 64 types of relationship annotations.

TABLE VIII
COMPARISON EXPERIMENTS FOR RS SCENE GRAPH GENERATION. BOLD INDICATES THE BEST PERFORMANCE AND UNDERLINE INDICATES THE SECOND BEST

	Recall Results			
Model	**R@20↑**	**R@50↑**	**R@100↑**	**R@500↑**
IMP[69]	17.2	24.1	30.5	45.8
Motifs[73]	31.2	39.3	44.7	52.0
VCTree[74]	31.4	39.6	44.9	52.0
Ringmo-SenseV2(ViTGCL)	**32.0**	**40.1**	**45.7**	**53.5**
	Mean Recall Results			
Model	**R@20↑**	**R@50↑**	**R@100↑**	**R@500↑**
IMP[69]	6.7	10.8	15.8	31.3
Motifs[73]	13.5	19.9	24.9	34.4
VCTree[74]	13.8	20.5	25.3	34.2
Ringmo-SenseV2(ViTGCL)	**16.4**	**23.0**	**28.3**	**37.0**

2) Implementation Details: All experiments are conducted using the PyTorch deep learning framework on a single NVIDIA GeForce RTX 3090 GPU with the batch size set to 2. We fine-tuned the model using an Adam optimizer with a momentum of 0.9 and a weight decay setting of 0, along with a cosine LR scheduler. The minimum LR was set to $1e^{-6}$, and the initial LR was set at $1e^{-3}$. For training rotation object detectors, we employ the ViTDet [70] with different initialization weights. The models for scene graph generation are evaluated using the SceneGraphBenchmark [71].

3) Comparison With the SOTA Methods: To validate the effectiveness of our foundational model in the RS scene graph generation task, we improve the SHA-GCL [74] method by utilizing ViT with pretrained Ringmo-SenseV2 and named it ViTGCL. SHA-GCL introduces a novel stacked hybrid-attention network, which facilitates both intramodal refinement and intermodal interaction, serving as the encoder. We conduct comparative experiments using different pretraining strategies under identical conditions with the ViTGCL. The results, as summarized in Table VIII, demonstrate that our model when initialized with parameters from the foundational pretraining. This highlights the improved performance of our foundational model in enhancing the remote sensing scene graph generation (RS-SGG) task.

The superior performance of the Ringmo-SenseV2 strategy could be attributed to its possibly more comprehensive feature learning and better generalization capabilities. This indicates that the foundation model's pretraining parameters, particularly in Ringmo-SenseV2, are effective at enhancing the model's ability to generalize from the training data to unseen data, which is crucial for tasks involving scene graph generation where context and detailed object relationships are significant.

4) Visualizations: Fig. 14 provides a visual representation of the results from the RS-SGG task. For visualization purposes, we partitioned the large RS image into four smaller sections, with the scene graph generation results partitioned accordingly. Notably, in part 4, the Ringmo-SenseV2 model demonstrates superior ability in extracting relationships

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

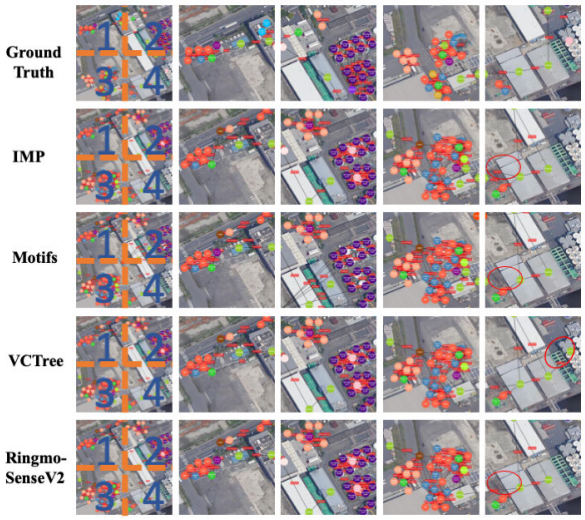


Fig. 14. Visualization results of RS scene graph generation comparative experiments.

between geographic features, accurately capturing close-to relationships between distant buildings, whereas other models do not perform as well in capturing these relationships. It is evident that our method accurately captures long-distance relationships between targets, outperforming other methods under the same experimental conditions. This highlights the advantage of the Ringmo-SenseV2 model.

V. Limitations

Despite the promising results demonstrated by Ringmo-SenseV2, there are several limitations. The model's performance is highly dependent on the quality and diversity of the training data, and its generalization capability may be limited in scenarios where the data distribution significantly deviates from the training data. Although we have designed the model with sophisticated components such as the MoHE Transformer and HT, the use of these complex modules requires significant computational resources, which may limit its application in resource-constrained environments. Additionally, the prediction accuracy may decrease in cases of low-resolution or sparse data, particularly when predicting small or fast-moving objects. Specific tasks such as urban expansion prediction, synthetic aperture radar (SAR) images, and hyperspectral data applications may require further adjustments to adapt to these different types of RS data, demanding stronger adaptability and generalization capabilities.

VI. Conclusion

This study introduces Ringmo-SenseV2, a universal predictive foundational model for multisource, heterogeneous, and time-series RS data. Ringmo-SenseV2 effectively adapts to feature representations in heterogeneous data through MoHE encoder and resolves complex dependency modeling challenges through the innovative introduction of HT. Furthermore, by introducing evolution speed factors and AM strategy, the model successfully adapts to diverse scenarios. Moreover, Ringmo-SenseV2 demonstrates highly competitive performance in multiple RS prediction tasks, offering a powerful tool for the analysis and prediction of time-series RS data. The outcomes of this study open new opportunities and horizons for various fields such as environmental protection and natural disaster prediction.

References

[1] A. Houenou, P. Bonnifait, V. Cherfaoui, and W. Yao, "Vehicle trajectory prediction based on motion model and maneuver recognition," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Nov. 2013, pp. 4363–4369.

[2] N. Deo and M. M. Trivedi, "Convolutional social pooling for vehicle trajectory prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 1468–1476.

[3] L. Lin, W. Li, H. Bi, and L. Qin, "Vehicle trajectory prediction using LSTMs with spatial–temporal attention mechanisms," *IEEE Intell. Transp. Syst. Mag.*, vol. 14, no. 2, pp. 197–208, Mar. 2022.

[4] A. H. Nielsen, A. Iosifidis, and H. Karstoft, "CloudCast: A satellite-based dataset and baseline for forecasting clouds," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3485–3494, 2021.

[5] C. Requena-Mesa, V. Benson, M. Reichstein, J. Runge, and J. Denzler, "EarthNet2021: A large-scale dataset and challenge for Earth surface forecasting as a guided video prediction task," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 1132–1142.

[6] X. Sun et al., "RingMo: A remote sensing foundation model with masked image modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2022, Art. no. 5612822.

[7] D. Wang et al., "Advancing plain vision transformer toward remote sensing foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2022, Art. no. 5607315.

[8] F. Yao et al., "RingMo-sense: Remote sensing foundation model for spatiotemporal prediction via spatiotemporal evolution disentangling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5620821.

[9] X. Shi, Z. Chen, H. Wang, D. Yeung, W. K. Wong, and W. Woo, "Convolutional LSTM network: A machine learning approach for precipitation nowcasting," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2015, pp. 802–810.

[10] X. Shi et al., "Deep learning for precipitation nowcasting: A benchmark and a new model," in *Proc. Adv. Neural Inf. Process. Syst.*, Jun. 2017, pp. 5622–5632.

[11] Y. Wang, M. Long, J. Wang, Z. Gao, and P. S. Yu, "PredRNN: Recurrent neural networks for predictive learning using spatiotemporal LSTMs," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Dec. 2017, pp. 879–888.

[12] Y. Wang, Z. Gao, M. Long, J. Wang, and P. S. Yu, "PredRNN++: Towards a resolution of the deep-in-time dilemma in spatiotemporal predictive learning," in *Proc. PMLR Int. Conf. Mach. Learn.*, Jul. 2018, pp. 5123–5132.

[13] Y. Wang et al., "PredRNN: A recurrent neural network for spatiotemporal predictive learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2208–2225, Feb. 2022.

[14] A. Graves and A. Graves, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997, doi: 10.1162/neco.1997.9.8.1735.

[15] F. Altché and A. de La Fortelle, "An LSTM network for highway trajectory prediction," in *Proc. IEEE 20th Int. Conf. Intell. Transport. Syst. (ITSC)*, Jun. 2017, pp. 353–359.

[16] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[17] C. Yu, X. Ma, J. Ren, H. Zhao, and S. Yi, "Spatio-temporal graph transformer networks for pedestrian trajectory prediction," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 507–523.

[18] A. Postnikov, A. Gamayunov, and G. Ferrer, "Transformer based trajectory prediction," 2021, *arXiv:2112.04350*.

[19] Z. Sui, Y. Zhou, X. Zhao, A. Chen, and Y. Ni, "Joint intention and trajectory prediction based on transformer," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Sep. 2021, pp. 7082–7088.

[20] W. Chen, F. Wang, and H. Sun, "S^{2}TNet: Spatio-temporal transformer networks for trajectory prediction in autonomous driving," in *Proc. Asian Conf. Mach. Learn.*, 2021, pp. 454–469.

[21] L. Achaji, T. Barry, T. Fouqueray, J. Moreau, F. Aioun, and F. Charpillet, "PreTR: Spatio-temporal non-autoregressive trajectory prediction transformer," in *Proc. IEEE 25th Int. Conf. Intell. Transp. Syst. (ITSC)*, Oct. 2022, pp. 2457–2464.

[22] L. Li, M. Pagnucco, and Y. Song, "Graph-based spatial transformer with memory replay for multi-future pedestrian trajectory prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 2231–2241.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.

[23] T. Zhan, Y. Song, J. Wang, and L. Wang, "VideoMAE: Masked autoencoders are data-efficient learners for self-supervised video pre-training," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 10078–10093.

[24] X. Liu et al., "M2IST: Multi-modal interactive side-tuning for efficient referring expression comprehension," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Mar. 2025.

[25] A. Arnab, M. Dehghani, G. Heigold, C. Sun, M. Lucic, and C. Schmid, "ViViT: A video vision transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6836–6846.

[26] Y. Cong et al., "SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 197–211.

[27] I. Bozcan and E. Kayacan, "AU-AIR: A multi-modal unmanned aerial vehicle dataset for low altitude traffic surveillance," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2020, pp. 8504–8510.

[28] L. Wen et al., "Detection, tracking, and counting meets drones in crowds: A benchmark," 2021, *arXiv:2105.02440*.

[29] P. Zhu et al., "Detection and tracking meet drones challenge," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7380–7399, Nov. 2022.

[30] Q. Yin et al., "Detecting and tracking small and dense moving objects in satellite videos: A benchmark," 2021, *arXiv:2111.12960*.

[31] Z. Liu et al., "Robust multi-drone multi-target tracking to resolve target occlusion: A benchmark," *IEEE Trans. Multimedia*, vol. 25, pp. 1462–1476, 2023.

[32] P. Zhu, T. Peng, D. Du, H. Yu, L. Zhang, and Q. Hu, "Graph regularized flow attention network for video animal counting from drones," *IEEE Trans. Image Process.*, vol. 30, pp. 5339–5351, 2021.

[33] N. Jiang et al., "Anti-UAV: A large multi-modal benchmark for UAV tracking," 2021, *arXiv:2101.08466*.

[34] D. Du et al., "The unmanned aerial vehicle benchmark: Object detection and tracking," 2018, *arXiv:1804.00518*.

[35] A. Robicquet, A. Sadeghian, A. Alahi, and S. Savarese, "Learning social etiquette: Human trajectory understanding in crowded scenes," in *Proc. 14th Eur. Conf. Comput. Vis. (ECCV)*, Amsterdam, The Netherlands. Cham, Switzerland: Springer, Jan. 2016, pp. 549–565.

[36] T. Shu, D. Xie, B. Rothrock, S. Todorovic, and S.-C. Zhu, "Joint inference of groups, events and human roles in aerial videos," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 4576–4584.

[37] C. Zhang et al., "WebUAV-3M: A benchmark for unveiling the power of million-scale deep UAV tracking," 2022, *arXiv:2201.07425*.

[38] M. Zhao, S. Li, S. Xuan, L. Kou, S. Gong, and Z. Zhou, "SatSOT: A benchmark dataset for satellite video single object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5617611.

[39] L. A. Varga, B. Kiefer, M. Messmer, and A. Zell, "SeaDronesSee: A maritime benchmark for detecting humans in open water," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 3686–3696.

[40] B. H. B. Hui et al., "A dataset for infrared image dim-small aircraft target detection and tracking under ground / air background," *Sci. Data Bank*, vol. 5, p. 12, Jul. 2022.

[41] A. Lerner, Y. Chrysanthou, and D. Lischinski, "Crowds by example," *Comput. Graph. Forum*, vol. 26, no. 3, pp. 655–664, Sep. 2007.

[42] S. Pellegrini, A. Ess, and L. Van Gool, "Improving data association by joint modeling of pedestrian trajectories and groupings," in *Computer Vision—ECCV 2010*. Berlin, Germany: Springer-Verlag, 2010, pp. 452–465.

[43] M.-F. Chang et al., "Argoverse: 3D tracking and forecasting with rich maps," 2019, *arXiv:1911.02620*.

[44] C. Requena-Mesa, V. Benson, M. Reichstein, J. Runge, and J. Denzler, "EarthNet2021: A large-scale dataset and challenge for Earth surface forecasting as a guided video prediction task," 2021, *arXiv:2104.10066*.

[45] W. Zhou, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, pp. 600–612, 2004.

[46] H. Bao, W. Wang, L. Dong, and F. Wei, "VL-BEiT: Generative vision-language pretraining," 2022, *arXiv:2206.01127*.

[47] H. Fan et al., "Multiscale vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6824–6835.

[48] Z. Liu et al., "Video Swin transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 3202–3211.

[49] N. Shazeer et al., "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," 2017, *arXiv:1701.06538*.

[50] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," 2021, *arXiv:2101.03961*.

[51] W. Wang, H. Bao, D. Li, and F. Wei, "VLMo: Unified vision-language pre-training with mixture-of-modality-experts," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2021, pp. 32897–32912.

[52] Y. Gao, Y. Feng, S. Ji, and R. Ji, "HGNN+: General hypergraph neural networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3181–3199, Mar. 2023.

[53] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2019, pp. 8–14.

[54] Z. Gao, C. Tan, L. Wu, and S. Z. Li, "SimVP: Simpler yet better video prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 3170–3180.

[55] L. Mou, Y. Hua, P. Jin, and X. X. Zhu, "ERA: A data set and deep learning benchmark for event recognition in aerial videos [software and data sets]," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 125–133, Dec. 2020.

[56] Y. Wang, J. Zhang, H. Zhu, M. Long, J. Wang, and P. S. Yu, "Memory in memory: A predictive neural network for learning higher-order non-stationarity from spatiotemporal dynamics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9146–9154.

[57] V. Le Guen and N. Thome, "Disentangling physical dynamics from unknown factors for unsupervised video prediction," 2020, *arXiv:2003.01460*.

[58] Y. Lyu, G. Vosselman, G.-S. Xia, A. Yilmaz, and M. Y. Yang, "UAVid: A semantic segmentation dataset for UAV imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 165, pp. 108–119, Jul. 2020.

[59] A. Van Etten, D. Hogan, J. M. Manso, J. Shermeyer, N. Weir, and R. Lewis, "The multi-temporal urban development SpaceNet dataset," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6398–6407.

[60] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, 2022.

[61] D. Simon, "Kalman filtering," *Embedded Syst. Program.*, vol. 14, no. 6, pp. 72–79, Jun. 2001.

[62] J. L. Elman, "Finding structure in time," *Cogn. Sci.*, vol. 14, no. 2, pp. 179–211, Mar. 1990.

[63] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.

[64] F. Giuliari, I. Hasan, M. Cristani, and F. Galasso, "Transformer networks for trajectory forecasting," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 10335–10342.

[65] Y. Yuan, X. Weng, Y. Ou, and K. Kitani, "AgentFormer: Agent-aware transformers for socio-temporal multi-agent forecasting," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9813–9823.

[66] F. Giuliari, I. Hasan, M. Cristani, and F. Galasso, "Transformer networks for trajectory forecasting," 2020, *arXiv:2003.08111*.

[67] M. Mueller, N. G. Smith, and B. Ghanem, "A benchmark and simulator for UAV tracking," in *Proc. 14th Eur. Conf. Comput. Vis. (ECCV)*. Amsterdam, The Netherlands. Cham, Switzerland: Springer, Oct. 2016, pp. 445–461.

[68] D. Xu, Y. Zhu, C. B. Choy, and L. Fei-Fei, "Scene graph generation by iterative message passing," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 5410–5419.

[69] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

[70] Y. Li, H. Mao, R. Girshick, and K. He, "Exploring plain vision transformer backbones for object detection," in *Proc. 17th Eur. Conf. Comput. Vis. (ECCV)*, Tel Aviv, Israel. Cham, Switzerland: Springer, Oct. 2022, pp. 280–296.

[71] K. Tang, Y. Niu, J. Huang, J. Shi, and H. Zhang, "Unbiased scene graph generation from biased training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3716–3725.

[72] R. Zellers, M. Yatskar, S. Thomson, and Y. Choi, "Neural motifs: Scene graph parsing with global context," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5831–5840.

[73] K. Tang, H. Zhang, B. Wu, W. Luo, and W. Liu, "Learning to compose dynamic tree structures for visual contexts," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6619–6628.

[74] X. Dong, T. Gan, X. Song, J. Wu, Y. Cheng, and L. Nie, "Stacked hybrid-attention and group collaborative learning for unbiased scene graph generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 19405–19414.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:23:08 UTC from IEEE Xplore. Restrictions apply.